

sdstudio

A Companion Package for Designing
and Managing surveydown Surveys

Pingfan Hu

George Washington University

GW OSCON 2026



About Me



- **Pingfan Hu**, PhD Candidate at George Washington University, advised by Dr. John Helveston.
- Research interests:
 1. **Data Science** in transportation systems and human behavior
 2. **Open-Source Software** in **R** tools for survey research
 3. **Agentic AI Engineering** around Claude Code
- For more information, visit pingfan.org

jhelvy.shinyapps.io

 **Welcome to the Ultimate Superhero Survey!** 

Thank you for participating in our research on superhero preferences and powers! This survey will help us understand what makes superheroes appealing to different people.

This survey should take about 3-5 minutes to complete.

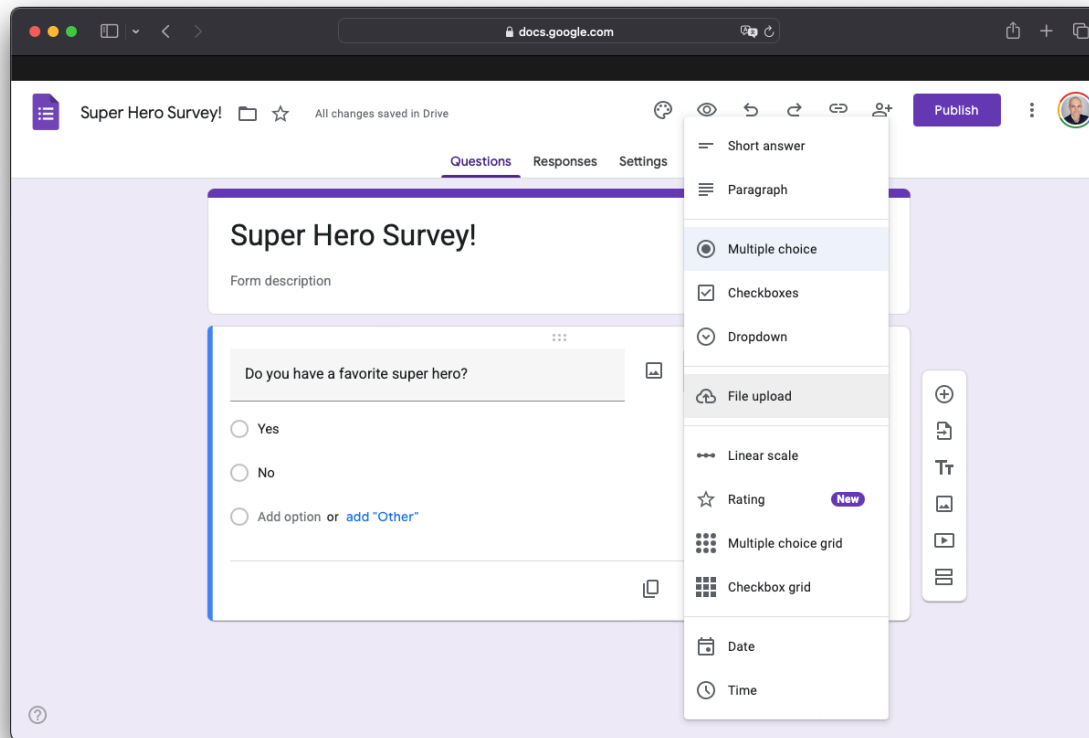
Do you have a favorite super hero? *

☐ Yes

☒ No

Next

WYSIWYG Interface



- ⊗ Reproducibility
- ⊗ Version control
- ⊗ Limited features
- ⊗ Open source

Why not make surveys
from **code**?

~~Limitations~~

Features

- ✓ Reproducibility
- ✓ Version control
- ✓ Lots of features
- ✓ Open source

Introducing **surveydown**!



Survey design using Quarto and Shiny



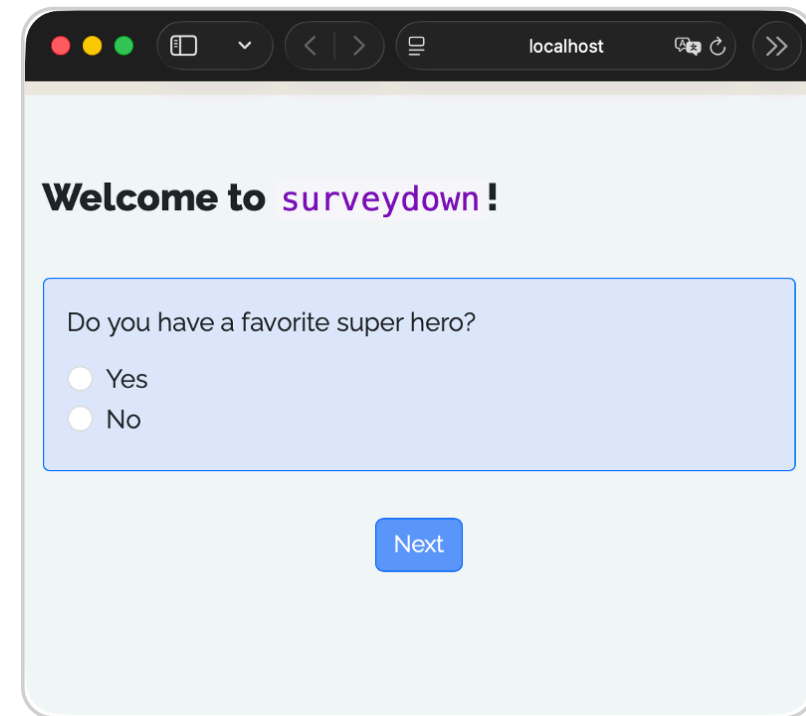
+



+



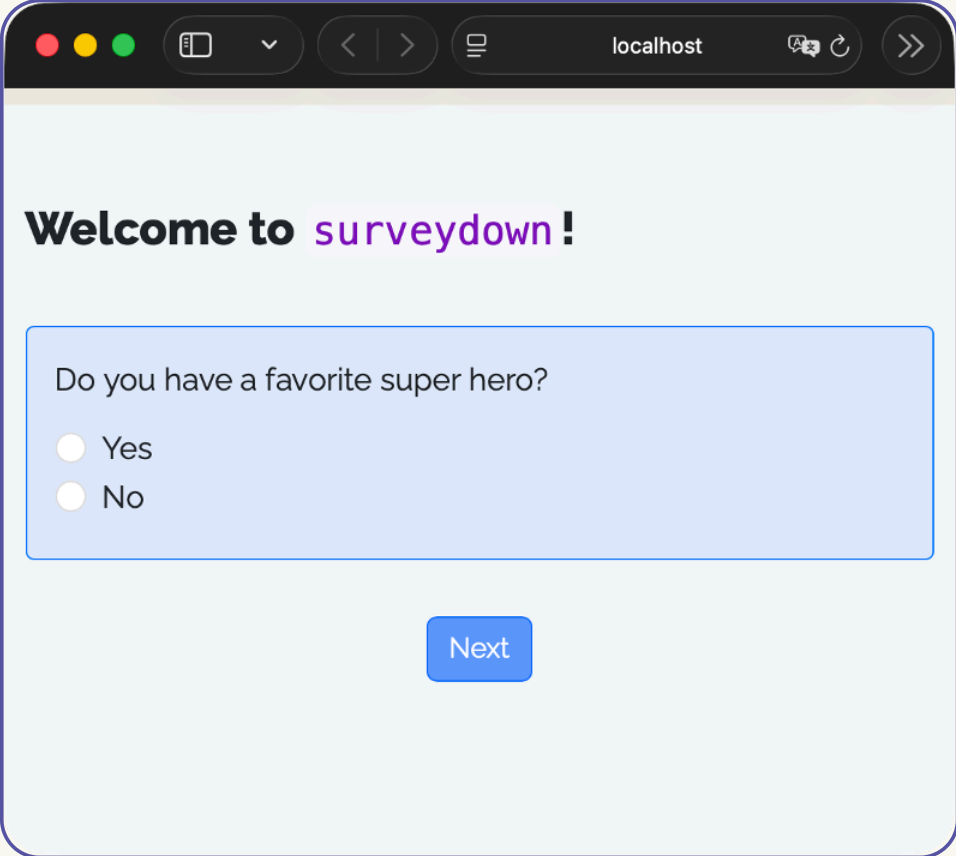
=



survey.qmd

```
1 ---
2 survey-settings:
3   use-cookies: true
4   rate-survey: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id   = "has_fav_hero",
19   label = "Do you have a favorite super hero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25 ```
26
27 --- some_other_page
28
29 Other content...
```

Rendered Survey



Welcome to surveydown!

Do you have a favorite super hero?

☐ Yes

☐ No

Next

But...



Code-Only Interface

No GUI. Not everyone is comfortable working purely in code.



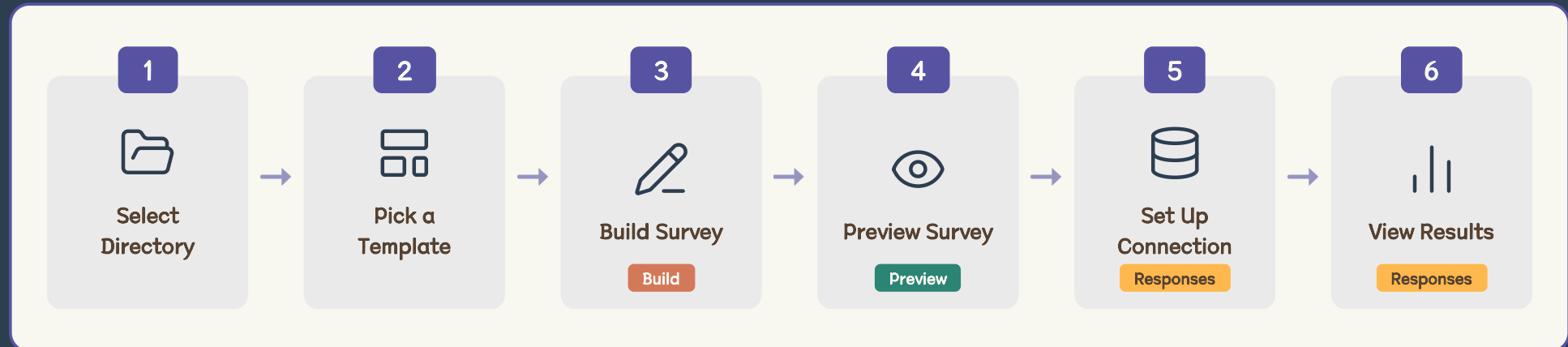
Learning Curve

The survey pipeline is not obvious. It takes time to learn.

Introducing **sdstudio**!



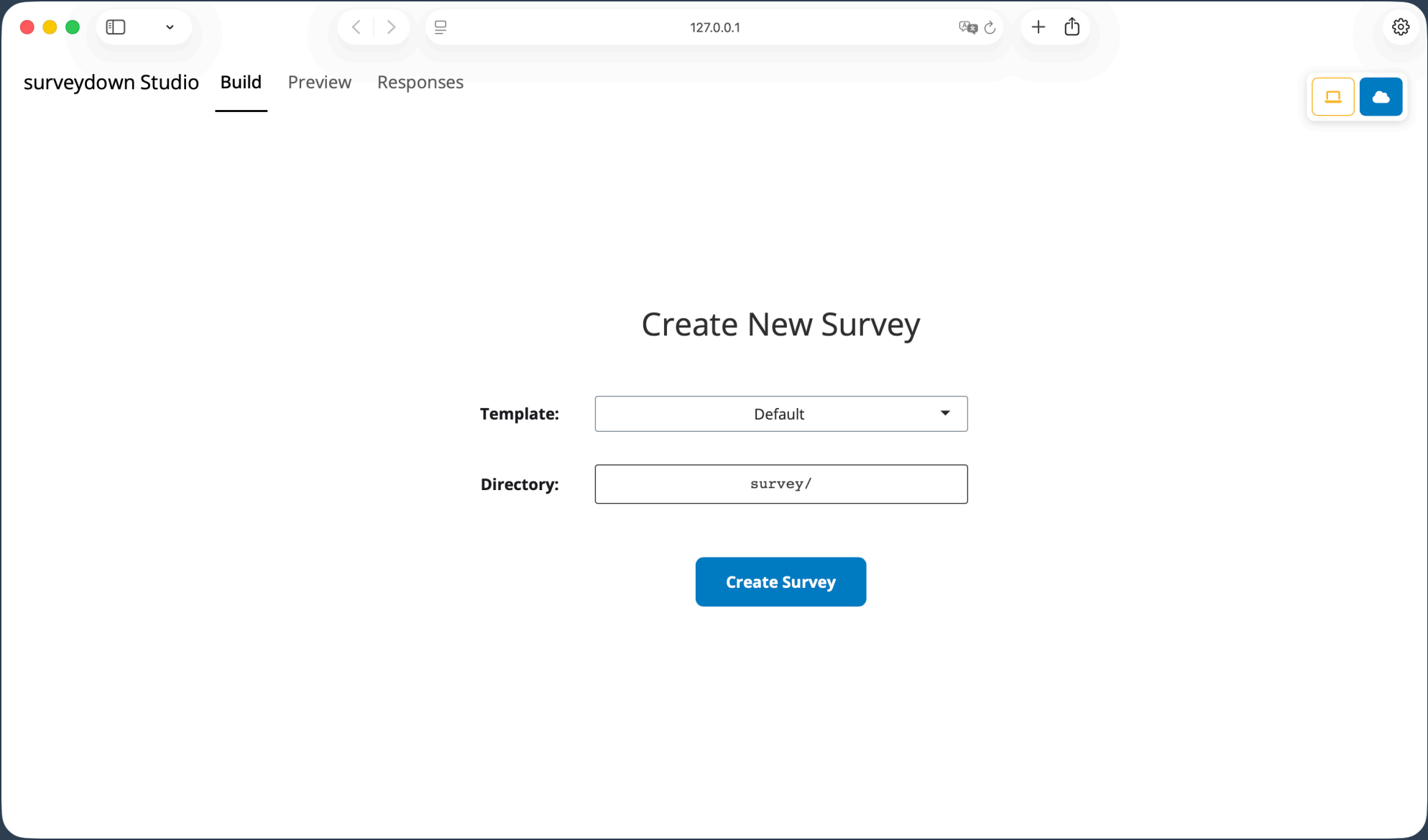
Studio Workflow



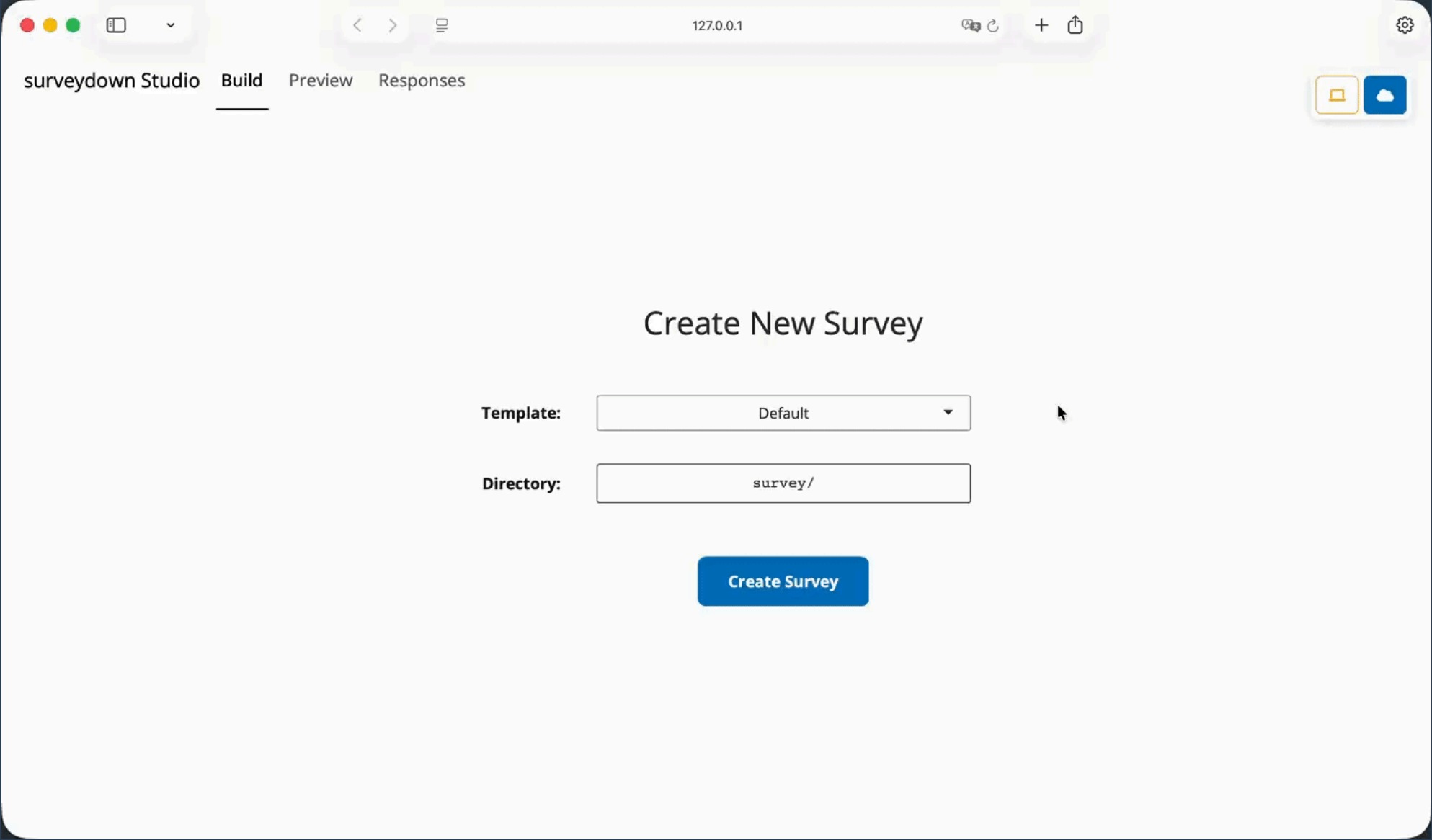
Launch with a simple command

```
sdstudio :: launch()
```

Landing Page



Start with a Template



The screenshot shows the 'surveydown Studio' application window. The title bar includes standard macOS window controls (red, yellow, green buttons) and a dropdown menu. The address bar shows the version '127.0.0.1'. The navigation bar has four tabs: 'surveydown Studio', 'Build' (which is active and underlined), 'Preview', and 'Responses'. On the right side of the navigation bar, there are icons for a folder, a cloud, and a settings gear. The main content area is titled 'Create New Survey' and contains two form fields. The first field is labeled 'Template:' and has a dropdown menu with 'Default' selected. The second field is labeled 'Directory:' and contains the text 'survey/'. Below these fields is a blue button labeled 'Create Survey'.

surveydown Studio Build Preview Responses

Create New Survey

Template: Default

Directory: survey/

Create Survey

surveydown.org/templates

surveydown.org

About

Documentation

Templates

Blog

FAQ

Templates

Basic

Default Template

Question Types

Questions Defined with YAML

Conditions

Conditional Showing

Conditional Skipping

Conditional Stopping

Randomization

Random Options

Random Options Predefined

Option Shuffling

Reactivity

Reactive Questions

Reactive drilldown

External Redirect

Live Polling

Conjoint

Conjoint Buttons

Conjoint Tables

Custom

Custom Leaflet Map

Custom Plotly Chart

Templates

To help you get started with surveydown, we created some templates that cover many of the features of the package. Feel free to use any as a starting point for your own surveys or as a reference for specific features.

Call `sd_create_survey()` with proper arguments to create your desired templates in your preferred directory. Refer to the [Start with a template](#) section for more details.

Default Template

Hi! This is a default template of a surveydown survey. It has two pages with one question on each page.

Please make your choice below.

Which type of penguin do you like the best?

Adelie

Chinstrap

Gentoo

You need to insert next buttons with `sd_next()` on each page. By default, it will go to the next page defined in the survey and file but you can also specify the page with the `next_page` argument set to the name of the page you want to go to next.

Next

Made with surveydown

Source Code

Default Template

BASIC

A minimum template for starting from scratch.

Template - Question Types

Hi! This is a showcase of different question types and question formatting that surveydown supports. Please make your choice below.

Where do you want to go to?

Take me to the page on Question Types (the next page)

Take me to the page on Question Formatting

Skip to the end

Next

Made with surveydown

Source Code

Question Types

BASIC

A template showcasing all built-in [question types](#).

Template - Questions Defined with YAML

This is a showcase of defining questions with a `questions.yml` file. The `sd_questions()` allows a `url` argument default to `"survey.yml"`. If you have such file in your project root folder you can use it to support your question definition. You can also change to other file name or directory and define questions like this.

```
sd_questions(
  url = "survey_questions.yml",
  url = "folder_name/1/1a_name.yml"
)
```

Please click on the "Next" button to see the demo questions.

Next

Made with surveydown

Source Code

Questions Defined with YAML

BASIC

A template showcasing all built-in [question types](#) using a YAML file.

Categories

All (17)

Basic (3)

Conditions (3)

Conjoint (2)

Custom (2)

Randomization (3)

Reactivity (4)

Site made with

and

quarto

Pingfan Hu - GW OSCON 2026

15 / 44

Choose Your Directory

The screenshot shows the 'surveydown Studio' application window. The title bar includes standard macOS window controls (red, yellow, green buttons) and a dropdown menu. The address bar shows the URL '127.0.0.1'. The navigation bar has four tabs: 'surveydown Studio', 'Build' (which is active and underlined), 'Preview', and 'Responses'. In the top right corner, there are icons for a folder, a cloud, and a settings gear. The main content area is titled 'Create New Survey'. It contains two form fields: 'Template:' with a dropdown menu currently set to 'Default', and 'Directory:' with a dark input field containing the text 'survey/'. A blue button labeled 'Create Survey' is positioned below these fields.

surveydown Studio Build Preview Responses

Create New Survey

Template: Default

Directory: survey/

Create Survey

The Build Tab

surveydown Studio Build Preview Responses

Structure

Page: welcome

T

Default Template Welcome to...

Q

penguins | mc

What's your favorite penguin?

T

There are several ways to...

Page: page2

T

This is another page in...

Q

silly_word | text

Write a silly word here:

T

On this page, we don't...

Page: end

T

End This is the...

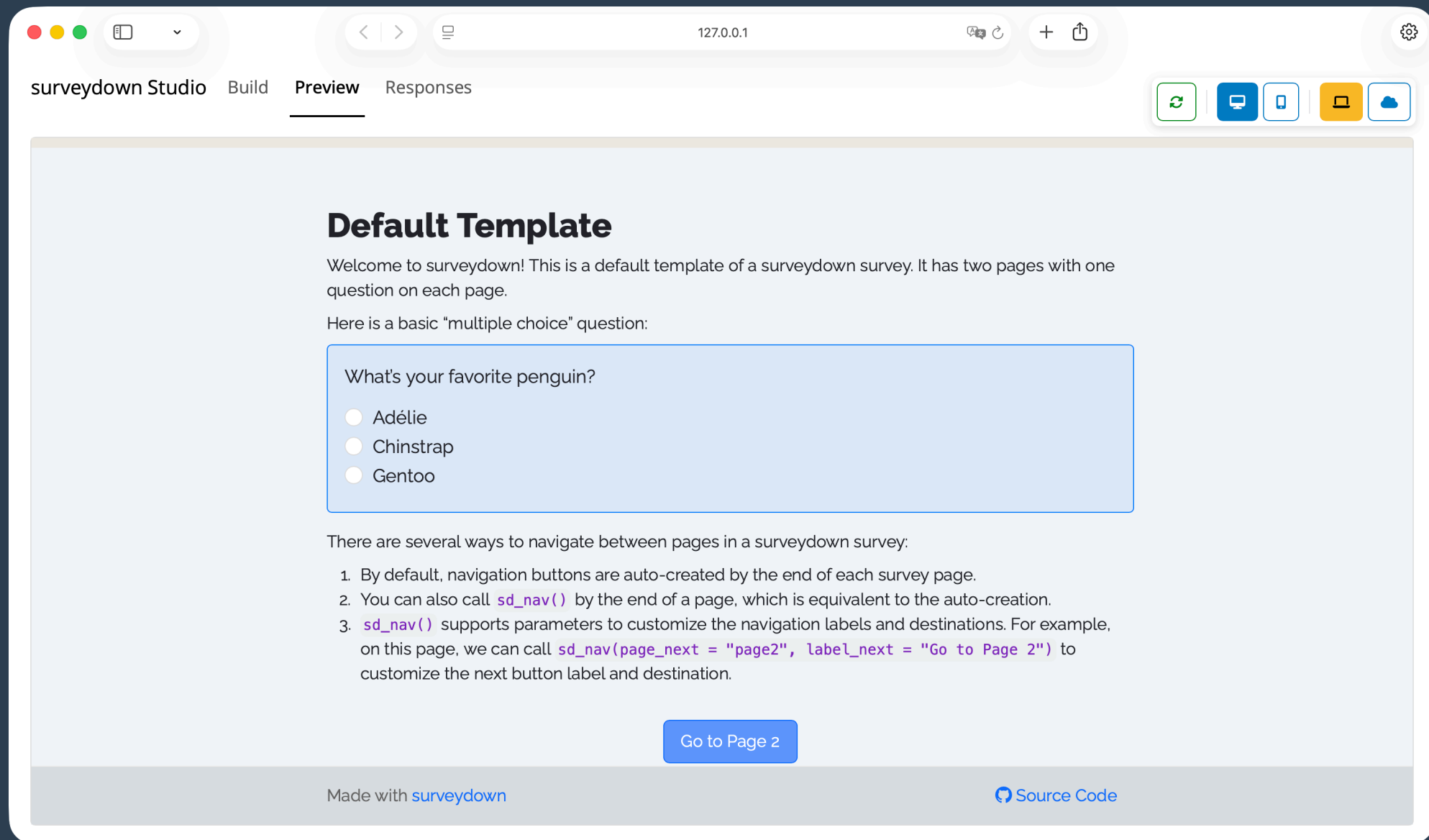
Add A New Page

Code

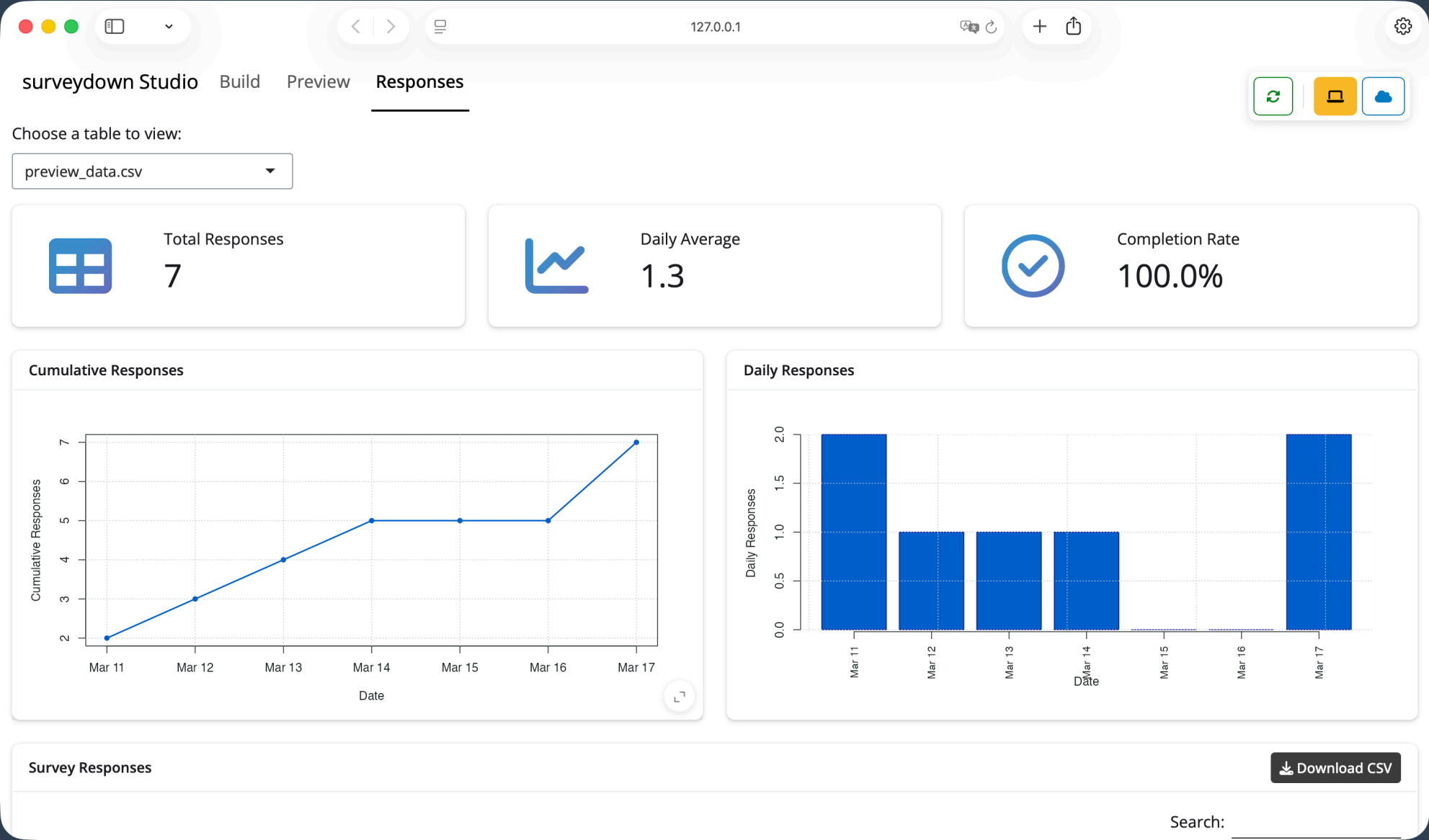
survey.qmd app.R

```
37 --- welcome
38
39 # Default Template
40
41 Welcome to surveydown! This is a default template of a surveydown survey. It has two
42 pages with one question on each page.
43
44 Here is a basic "multiple choice" question:
45
46 ```{r}
47 sd_question(
48   type = 'mc',
49   id   = 'penguins',
50   label = "What's your favorite penguin?",
51   option = c(
52     'Adélie' = 'adelie',
53     'Chinstrap' = 'chinstrap',
54     'Gentoo' = 'gentoo'
55   )
56 )
57
58 There are several ways to navigate between pages in a surveydown survey:
59
60 1. By default, navigation buttons are auto-created by the end of each survey page.
61 2. You can also call `sd_nav()` by the end of a page, which is equivalent to the
62    auto-creation.
63 3. `sd_nav()` supports parameters to customize the navigation labels and destinations.
64    For example, on this page, we can call `sd_nav(page_next = "page2", label_next = "Go to
65    Page 2")` to customize the next button label and destination.
66
67 ```{r}
```

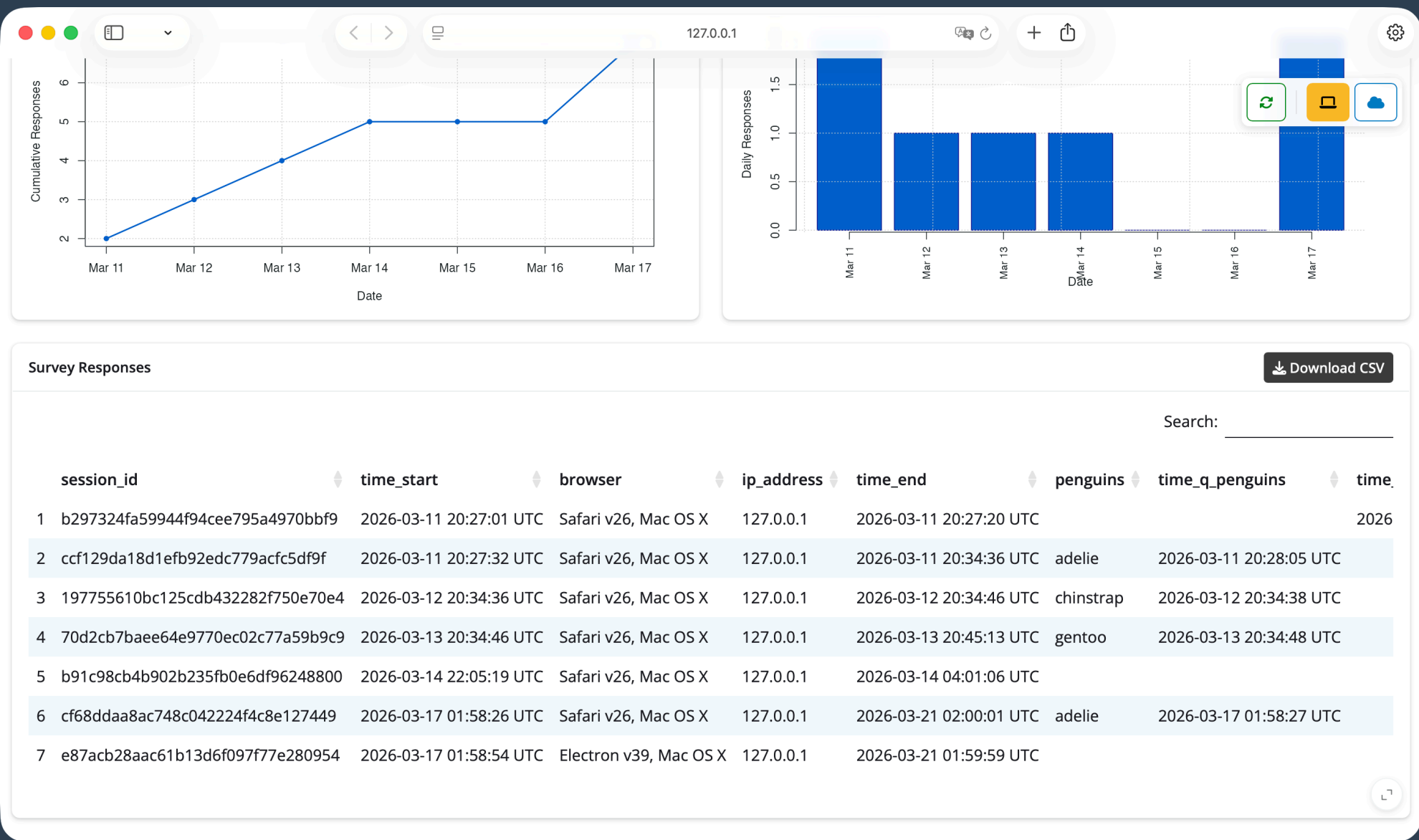

The Preview Tab



The Responses Tab

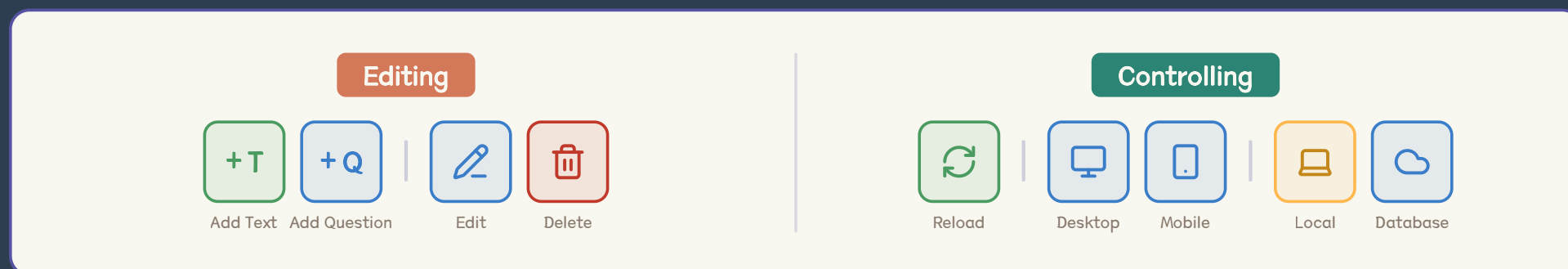


The Responses Tab - Continued

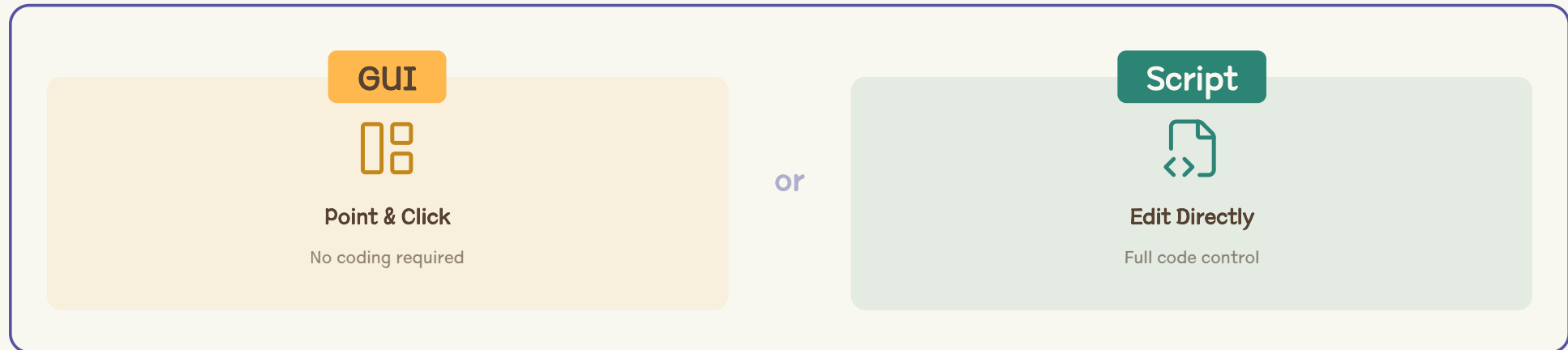


Studio Features

Self-explanatory Icons



Editing



GUI - Toggling

The screenshot displays the surveydown Studio application interface. At the top, there's a navigation bar with tabs for 'surveydown Studio', 'Build', 'Preview', and 'Responses'. Below this, the main workspace is divided into two panels: 'Structure' on the left and 'Code' on the right.

Structure Panel: This panel shows a hierarchical view of the survey structure. It includes a 'Page: page1' section with a text element ('# Title Some texts.') and a multiple-choice question ('penguins | mc' with the label 'What's your favorite penguin?'). Below this is a 'Page: end' section with a text element ('## End This is the...'). At the bottom of the Structure panel is an orange button labeled 'Add A New Page'.

Code Panel: This panel shows the R code for the survey. The code is organized into sections corresponding to the Structure panel. It starts with a library call for 'surveydown', followed by a page definition for 'page1' containing a title and a multiple-choice question. The code ends with a closing tag for the page and a final text element.

```
33 library(surveydown)
34 - `
35
36
37 --- page1
38
39 # Title
40
41 Some texts.
42
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id   = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 - `
55
56 --- end
57
58 ## End
59
60 This is the last page in the survey.
61
62 ```{r}
63 sd_close()
64 - `
```

GUI - Content Editing

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- At the top, there are tabs for "surveydown Studio", "Build", "Preview", and "Responses".
- Below the tabs, there is a "Structure" section with a list of pages and questions.
- The first page is "Page: page1". It contains a question titled "# Title Some texts." with a type of "mc" (multiple choice). The question text is "What's your favorite penguin?".
- The second page is "Page: end". It contains a question titled "## End This is the..." with a type of "mc".
- At the bottom of the Structure panel, there is a button labeled "Add A New Page".

Code Panel:

- The Code panel shows the R code for the survey. It is divided into two tabs: "survey.qmd" and "app.R".
- The "survey.qmd" tab is active, showing the following code:

```
33 library(surveydown)
34 ---
35
36
37 --- page1
38
39 # Title
40
41 Some texts.
42
43 {r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ---
55
56 --- end
57
58 ## End
59
60 This is the last page in the survey.
61
62 {r}
63 sd_close()
64 ---
```


GUI - Content Addition

The screenshot displays the surveydown Studio interface. The top navigation bar includes 'surveydown Studio', 'Build', 'Preview', and 'Responses'. The 'Structure' panel on the left shows a hierarchical view of the survey content, including 'Page: page1' with a title and a multiple-choice question about penguins, and 'Page: end' with a closing text. The 'Code' panel on the right shows the corresponding R code for the survey, including the 'survey.qmd' file and the 'app.R' script. The code defines the survey structure, including the title, question, and options.

Structure Panel:

- Page: page1
 - # Title New texts.
 - penguins | mc
 - What's your favorite penguin?
- Page: end
 - ## End This is the...

Code Panel:

```
33 library(surveydown)
34 ---
35
36
37 --- page1
38
39 # Title
40
41 New texts.
42
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id   = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ---
55
56 --- end
57
58 ## End
59
60 This is the last page in the survey.
61
62 ```{r}
63 sd_close()
64 ---
```

GUI - Page Addition

The screenshot displays the surveydown Studio interface. The top navigation bar includes 'surveydown Studio', 'Build', 'Preview', and 'Responses'. The 'Structure' panel on the left shows a hierarchical view of the survey pages. It lists 'Page: page1' with three items: a title '# Title New texts.', a multiple-choice question 'penguins | mc' with the label 'What's your favorite penguin?', and a text question 'text_question | text' with the label 'Write something:'. Below this is 'Page: end' with a final text item '## End This is the...'. An orange button at the bottom of the Structure panel says 'Add A New Page'. The 'Code' panel on the right shows the R code for the survey. It includes a 'new texts.' block with a multiple-choice question and a 'text_question' block with a text question. The code is as follows:

```
41 new texts.  
42  
43- ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id   = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54- ```  
55- ```{r}  
56 sd_question(  
57   type = "text",  
58   id   = "text_question",  
59   label = "Write something:"  
60 )  
61- ```  
62  
63  
64 --- end  
65  
66- ## End  
67  
68 This is the last page in the survey.  
69  
70- ```{r}  
71 sd_close()  
72- ```
```

GUI - Page Relocation

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- Page: page1**
 - # Title New texts.**
 - penguins | mc**
What's your favorite penguin?
 - text_question | text**
Write something:
- Page: page2**
 - This is page 2.**
- Page: end**
 - ## End This is the...**

At the bottom of the Structure panel is a button labeled "Add A New Page".

Code Panel:

The Code panel shows the R code for the survey, with tabs for "survey.qmd" and "app.R". The code is as follows:

```
41 new texts.  
42  
43- ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id   = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54- ```  
55- ```{r}  
56 sd_question(  
57   type = "text",  
58   id   = "text_question",  
59   label = "Write something:"  
60 )  
61- ```  
62  
63  
64 --- page2  
65  
66 This is page 2.  
67  
68  
69 --- end  
70  
71- ## End  
72
```

GUI - Content Relocation

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- Page: page1**
 - # Title New texts.** (Text question)
 - penguins | mc** (Multiple choice question: "What's your favorite penguin?")
 - text_question | text** (Text question: "Write something:")
- Page: page2**
 - This is page 2.** (Text question)
- Page: end**
 - ## End This is the...** (Text question)

At the bottom of the Structure panel is a button labeled "Add A New Page".

Code Panel:

The Code panel shows the R code for the survey, with tabs for `survey.qmd` and `app.R`. The `survey.qmd` tab is active, displaying the following code:

```
41 new texts.  
42  
43- ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id   = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54- ```  
55- ```{r}  
56 sd_question(  
57   type = "text",  
58   id   = "text_question",  
59   label = "Write something:"  
60 )  
61- ```  
62  
63  
64 --- page2  
65  
66 This is page 2.  
67  
68  
69 --- end  
70  
71- ## End  
72
```

Script Editing

The screenshot displays the surveydown Studio application interface. At the top, there are navigation tabs: "surveydown Studio", "Build", "Preview", and "Responses". Below these, the "Structure" panel on the left shows a hierarchical view of the survey pages. It includes "Page: page1" with a title question and a multiple-choice question about favorite penguins, "Page: page2" with a text question, and "Page: end" with a closing text. An "Add A New Page" button is at the bottom of the Structure panel. The "Code" panel on the right shows the R script for the survey, with tabs for "survey.qmd" and "app.R". The script defines two pages: "page1" with a multiple-choice question and "page2" with a text question. The script ends with a closing command.

Structure Panel:

- Page: page1
 - # Title New texts.
 - penguins | mc
What's your favorite penguin?
- Page: page2
 - This is page 2.
 - text_question | text
Write something:
- Page: end
 - ## End This is the...

Code Panel (survey.qmd):

```
44 sd_question(  
45   type = 'mc',  
46   id   = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54 ---  
55 --- page2  
56  
57 This is page 2.  
58  
59 {r}  
60 sd_question(  
61   type = "text",  
62   id   = "text_question",  
63   label = "Write something:"  
64 )  
65 ---  
66  
67 --- end  
68  
69 ## End  
70  
71 This is the last page in the survey.  
72  
73 {r}  
74 sd_close()  
75 ---
```


Instant Preview

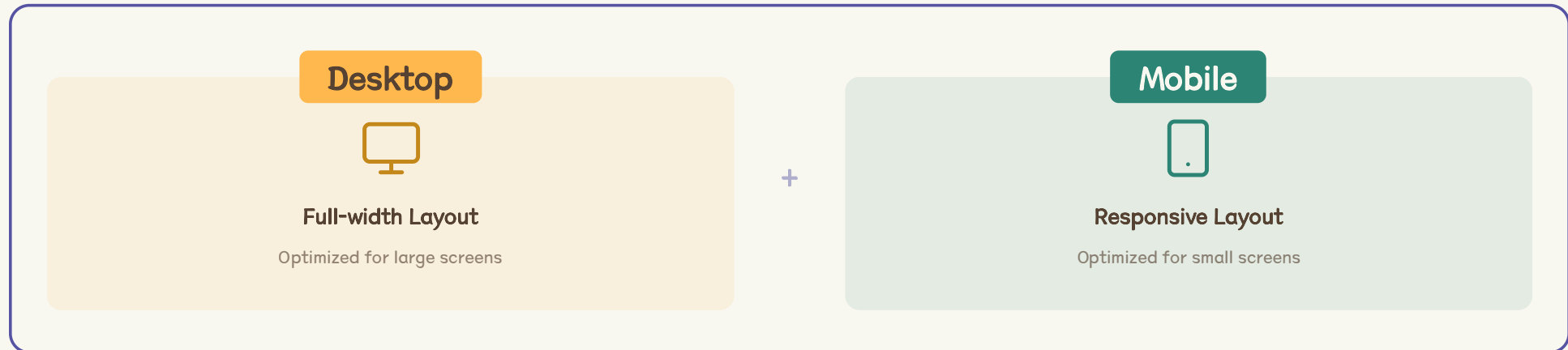
The screenshot displays the surveydown Studio application interface. At the top, there's a navigation bar with tabs for 'surveydown Studio', 'Build', 'Preview', and 'Responses'. The 'Build' tab is currently active. Below the navigation bar, the interface is split into two main panels: 'Structure' on the left and 'Code' on the right.

Structure Panel: This panel shows a hierarchical view of the survey structure. It includes three pages: 'Page: page1', 'Page: page2', and 'Page: end'. Each page contains a list of elements (questions and text blocks) with icons for adding, editing, and deleting them. For example, 'Page: page1' contains a title 'New texts.' and a multiple-choice question 'What's your favorite penguin?'. 'Page: page2' contains a text block 'This is page 2.' and a text question 'Write something:'. At the bottom of the Structure panel, there is a button labeled 'Add A New Page'.

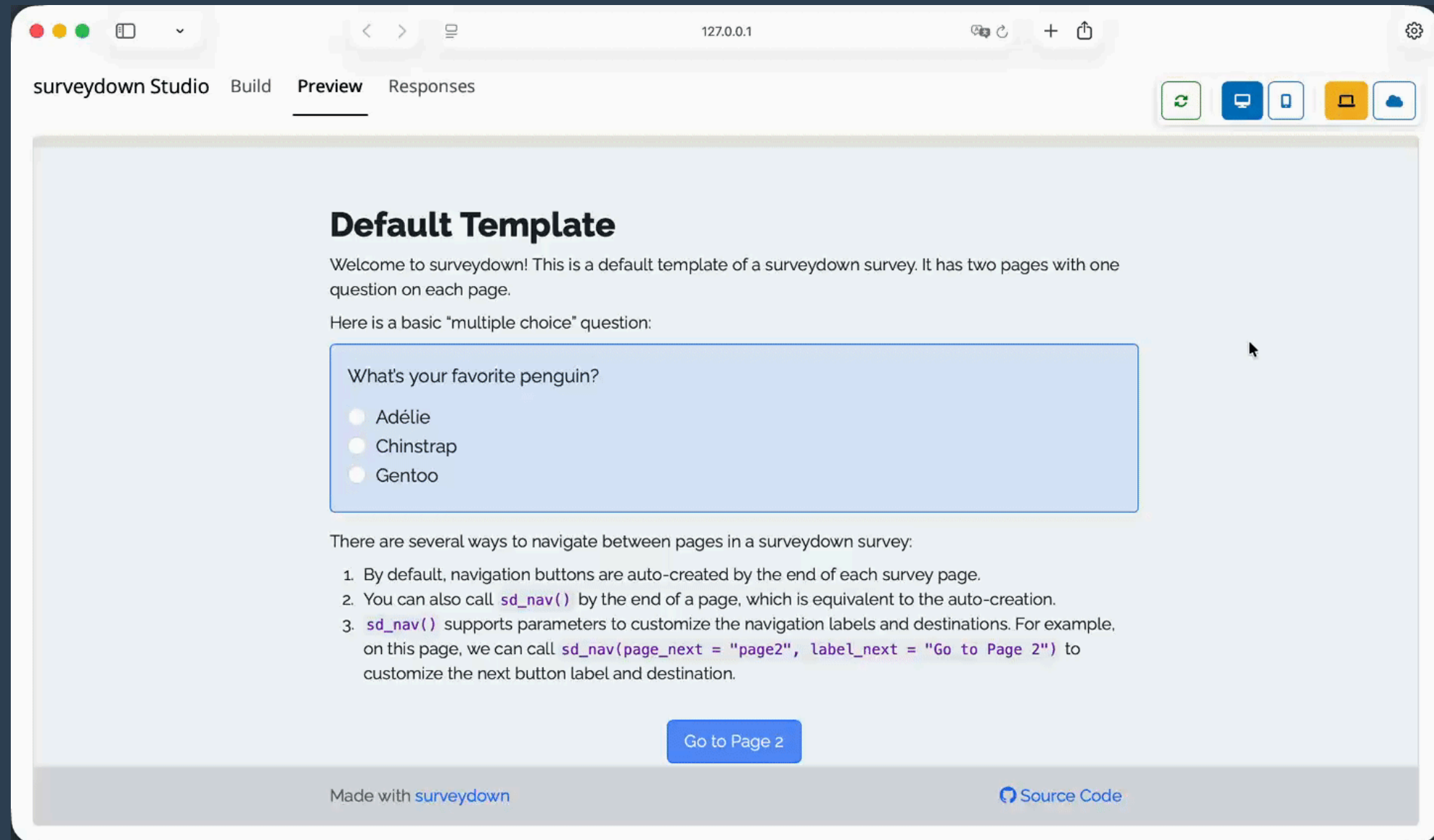
Code Panel: This panel shows the R code that generates the survey. It includes a file explorer at the top with 'survey.qmd' and 'app.R'. The code is written in R and defines the survey structure using the 'sd_question' function. The code is as follows:

```
34- ---
35-
36-
37- --- page1
38- # Title
39- New texts.
40-
41- {r}
42- sd_question(
43-   type = 'mc',
44-   id = 'penguins',
45-   label = "What's your favorite penguin?",
46-   option = c(
47-     'Adélie' = 'adelie',
48-     'Chinstrap' = 'chinstrap',
49-     'Gentoo' = 'gentoo'
50-   )
51- )
52- }
53- ---
54-
55- --- page2
56- This is page 2.
57-
58- {r}
59- sd_question(
60-   type = "text",
61-   id = "text_question",
62-   label = "Write something:"
63- )
64- }
```

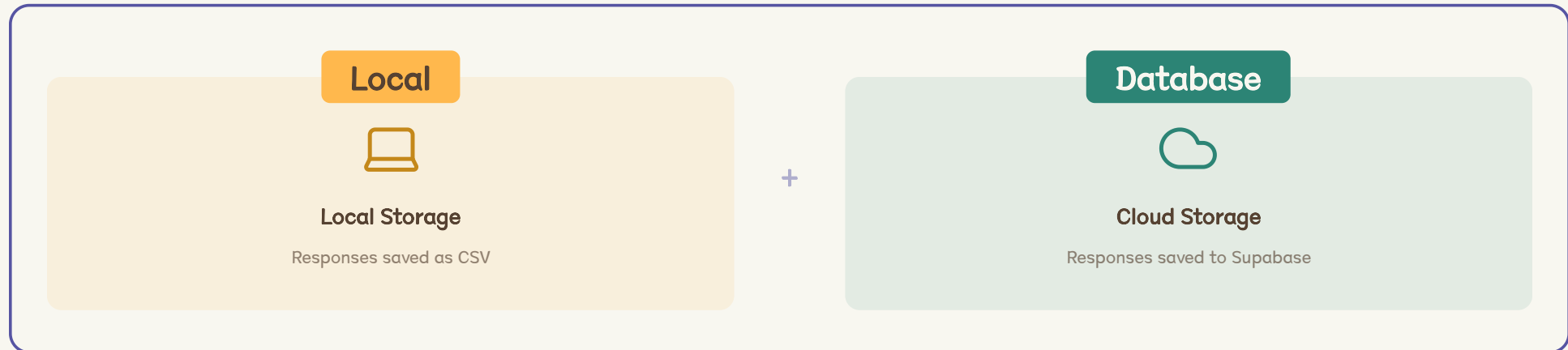
Views



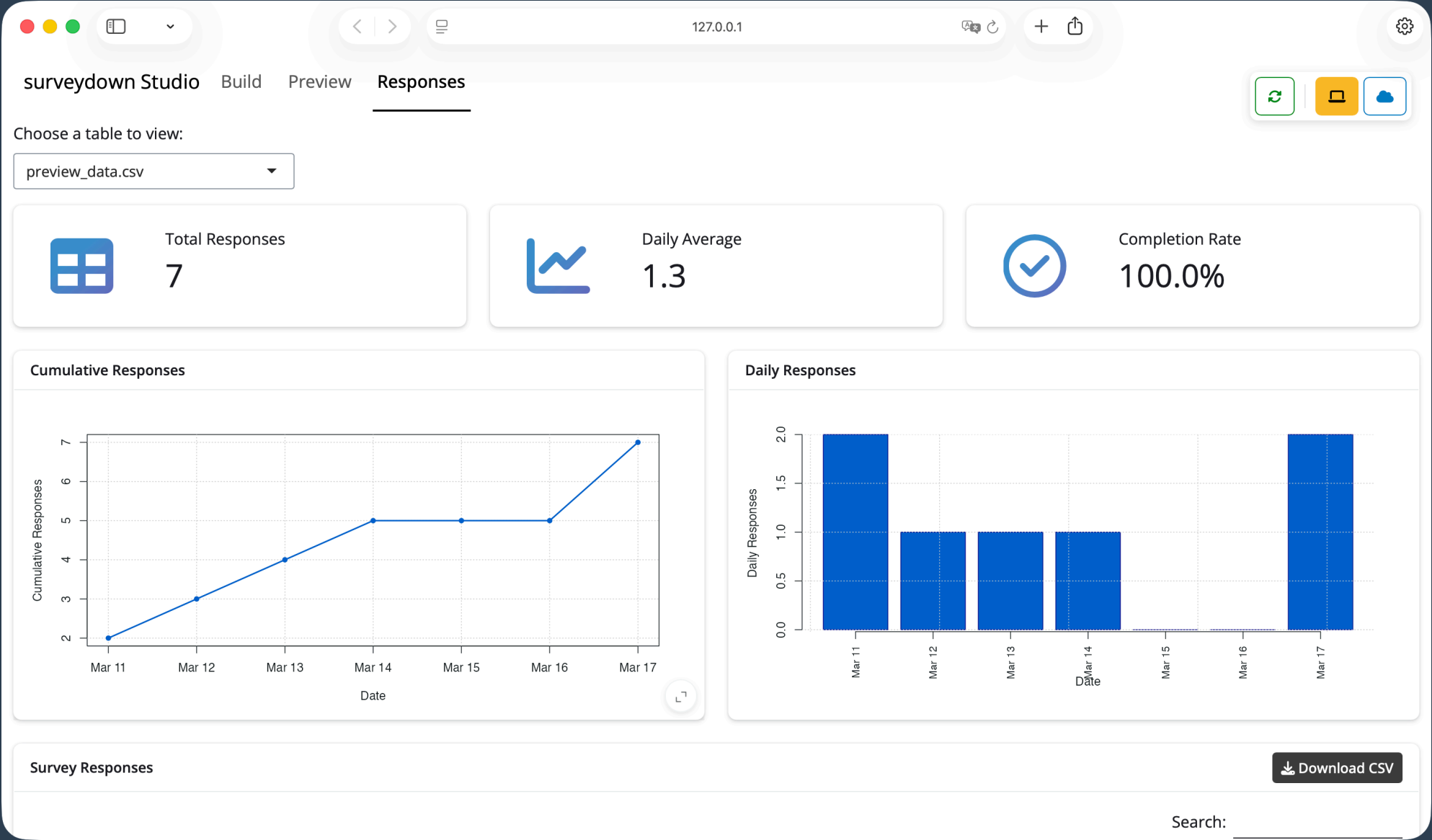
Desktop and Mobile Views



Responses



Local Responses



Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Settings

Not connected

Choose a table to view:

Select or type table name...

Total Responses

Daily Average

Completion Rate

Cumulative Responses

Daily Responses

Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Settings

Not connected

Choose a table to view:

Select or type table name...

Total Responses

Daily Average

Completion Rate

Cumulative Responses

Daily Responses

Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

× Close

● Not connected

Host:

Port:

Database Name:

postgres

User:

Password:

.....

Show

Table:

default

Test Connection

Choose a table to view:

Select or type table name... ▼

Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Close

Connected (GSSAPI disabled)

Host:

Port:

Database Name:

postgres

User:

Password:

.....

Show

Table:

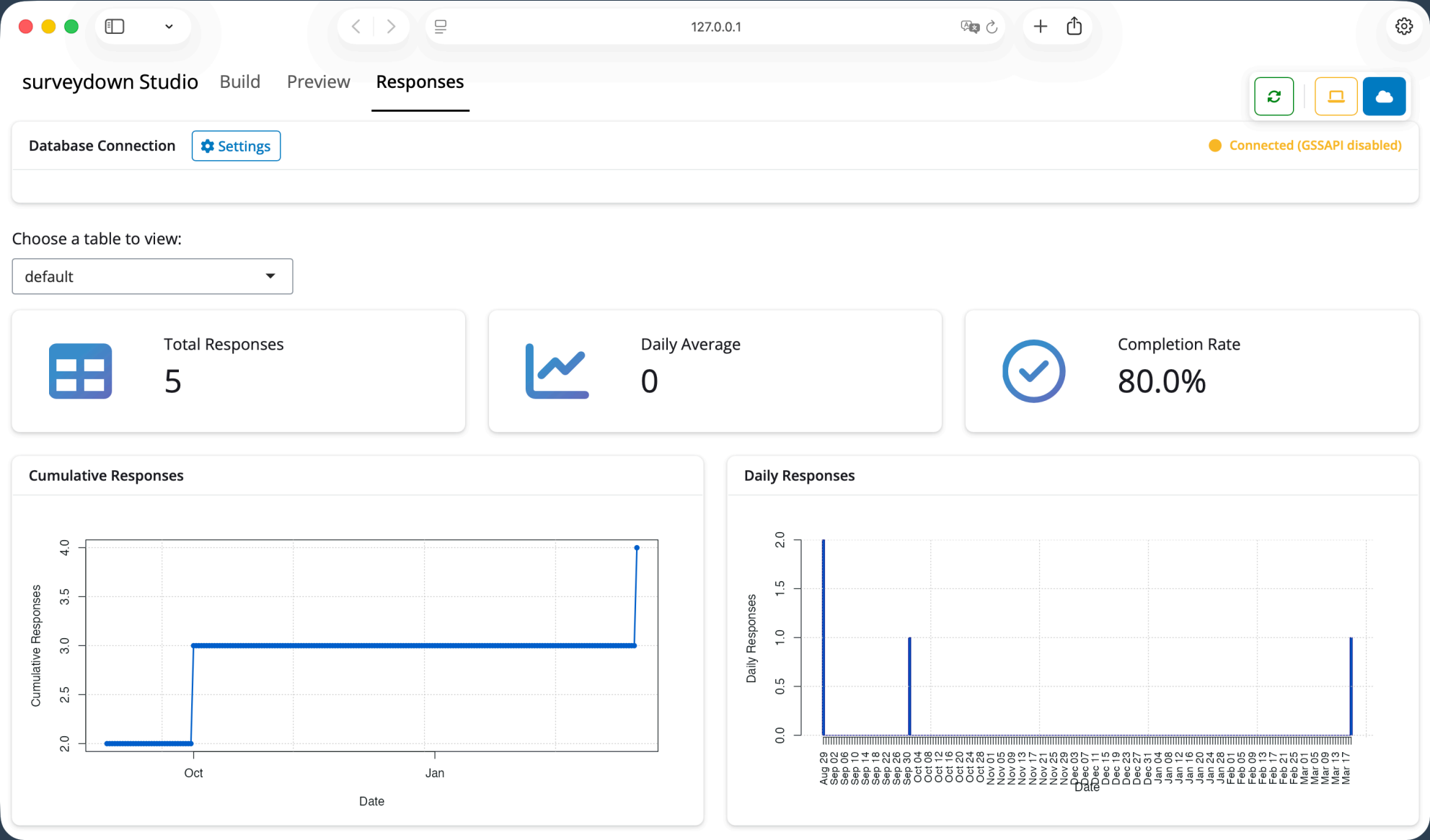
default

Test Connection

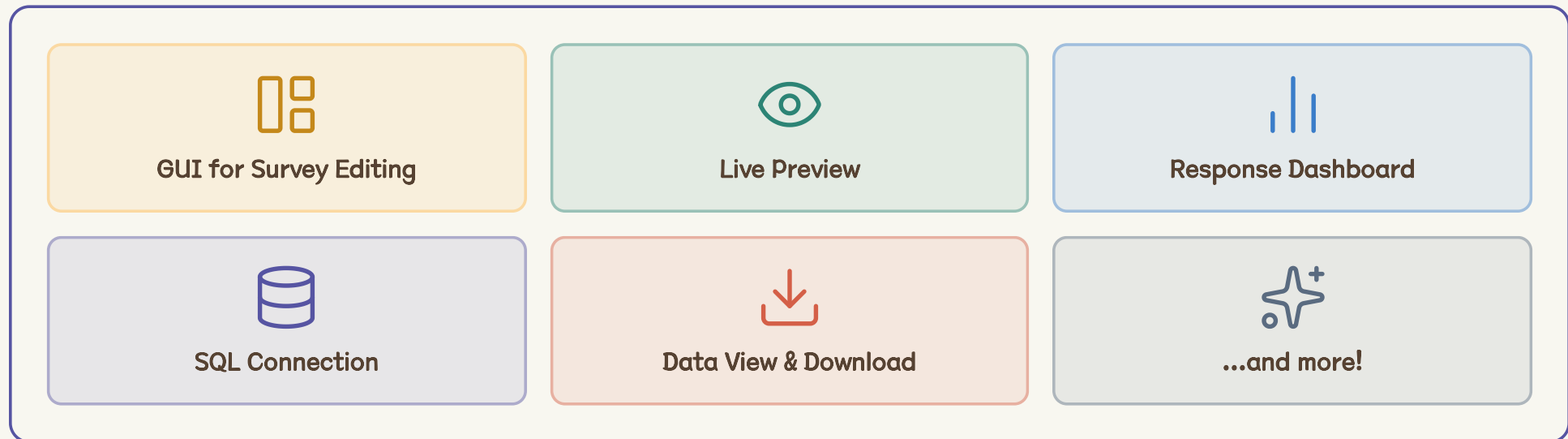
Choose a table to view:

default

Database Responses



sdstudio gives you...



surveydown + sdstudio

surveydown

 Text-based editing

 Published paper

 8000+ CRAN downloads

sdstudio

 GUI editing & live preview

 Actively developed

Open to contributions

 Better together!

Ways to help:

github.com/surveydown-dev/sdstudio/

- Try it out!
- Give it a ★
- Post an issue
- Join the GitHub discussion
- Contribute new features



Meet the team!



John Helveston, Ph.D.



Pingfan Hu



Bogdan Bunea

surveydown.org

